Aakash Kumar

AI Product Engineer | Applied AI, RAG & Semantic Search | Former CTO (Acquired Startup)

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SUMMARY

AI product engineer with 4+ years of building and shipping products from zero to scale. Former CTO of CloneMyTrips (acquired) and Head of AI at WanderOn. Builds production AI systems: semantic search and reranking over ~26M patent families, RAG pipelines, AI agents and workflow automation. Strong full-stack foundation (Python, TypeScript, React, Node.js) and a security background: top 5 globally on HackTheBox, 2x ETHGlobal winner, and holder of an Australian patent in ML-based vulnerability detection.

EXPERIENCE

AI Consultant | Shifu Ventures (Product Studio & VC)

Aug 2026 – Present | India

- Designed and built the core AI of **Photon Legal's Product Pulse**: a prior-art search engine over **~26M deduplicated Google Patents families**.
- Embedded patent text with **MiniLM** sentence embeddings for semantic vector retrieval and reranked candidates with **Cohere Rerank** to surface the closest matching prior art.
- Built a CI-gated evaluation harness that scores retrieval recall against the prior art patent examiners actually cited; found and fixed a family-level scoring flaw that had been understating recall and reversed a planned switch to a smaller corpus.
- Built an idea-scoring system that compares inventors' disclosures against matched prior art, helping client companies decide which inventions to file as patents.
- Work with Shifu Ventures portfolio startups on product, AI and engineering to get their products working reliably in production.

Head of AI | WanderOn

Oct 2025 – Jul 2026 | Gurugram, India

- Led AI across the organization; built internal AI tools that improved operational efficiency and launched AI agents to improve customer experience.
- Built AI-first systems for operations automation and product intelligence using LLM applications, RAG and multi-agent workflows. The company raised an INR 54 Cr Series A during this period.

Chief Technology Officer | CloneMyTrips (Acquired)

Mar 2024 – Sep 2025 | Bengaluru, India (Hybrid)

- Designed the platform architecture from scratch and led engineering and product through to the company's acquisition.
- Launched the **Dora AI Engine** in the mobile app, generating AI-curated travel itineraries that connect travel creators with their audiences.
- Worked directly with the founders on technical strategy, product roadmap and scaling.

Senior Frontend Developer (Web3) | Safle

Feb 2023 – Mar 2024 | Abu Dhabi, UAE (Remote)

- Built a crypto wallet that works without an internet connection, for a Web3 wallet and browser used by a global user base.

Co-Founder | Calmverse

Aug 2022 – Feb 2023 | Remote

- Co-founded a Web3 metaverse for mental health; owned the product concept and technical architecture.

Senior Frontend Developer | QuickSell

Jun 2022 – Nov 2022 | Mumbai, India

- Promoted from junior to senior developer; built React.js applications for a conversational commerce platform.

Lead Developer, Megham Cloud (IaaS) | Centre of Innovation & Incubation, NIT Warangal

Aug 2020 – Apr 2022 | Warangal, India

- Led development of the institute's Megham cloud infrastructure platform.

SELECTED CONSULTING & CLIENT WORK

- **Fintech & Web3 products** (Frontline BNPL, PayoutRails stablecoin payroll, HashFast payment links, NanoBond bond launchpad): built smart-contract and payment infrastructure, batch payouts and real-time settlement.
- **Nihaan Energy** (EV charging): built the NEO mobile app and web charger-management platform with real-time monitoring and OCPP compliance.
- 20+ products shipped across AI, fintech, Web3, consumer apps and clean energy.

PROJECTS

- **Wedding Album AI Shortlister**: picks the best photos from thousands using CLIP embeddings, aesthetic scoring, clustering-based duplicate removal and LLM curation. Launched on Product Hunt (Mar 2026). Python, CLIP, LLMs.
- **OKX DeFi AI Copilot**: AI agent that turns plain-English trading strategies into one-click swaps on Solana via the OKX DEX API. React 19, TypeScript, Perplexity API.
- **SafeLease**: privacy-preserving property leasing with zero-knowledge identity (Self Protocol) and four Solidity smart contracts on Celo.

SKILLS

AI / ML: LLM applications, RAG, semantic search, vector embeddings (MiniLM, CLIP), reranking (Cohere), AI agents, multi-agent systems, prompt engineering, OpenAI and Claude APIs, MCP

Languages: Python, TypeScript, JavaScript, SQL, Solidity, Bash

Backend & Data: Node.js, Express, Django, REST, GraphQL, PostgreSQL, MongoDB, Redis

Frontend: React, Next.js, React Native, Tailwind CSS

Cloud & DevOps: AWS, Docker, CI/CD, Vercel, Netlify, Linux, Git

Web3 & Security: Solana, Ethereum, Hedera, Celo, penetration testing, OWASP Top 10, threat modeling

EDUCATION

B.Tech, Computer Science and Engineering | National Institute of Technology (NIT), Warangal
2018 – 2022

ACHIEVEMENTS

- **Patent:** Australian Patent AU 2021105619, an ML-based method for identifying duplicate security vulnerabilities.
- **2x ETHGlobal hackathon winner** (ETHGlobal Singapore and Bangkok).
- **HackTheBox Omniscient rank**: top 5 globally among 400,000+ hackers.
- Earned INR 2 lakh+ in bug bounties for responsibly disclosing vulnerabilities to Dell, Samsung and others.
- Invited InfoSec speaker at the E&ICT Academy, NIT Warangal Faculty Development Program.
- Co-founded BugClaim (crowdsourced security SaaS) and CybSec NITW; organized PhantomCTF, NIT Warangal's first CTF.